



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	QUANTUM COMPUTATION AND QUANTUM INFORMATION				
Type of Course	Major Elective				
Semester	VIII				
Academic Level	400 - 499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	1. Linear Algebra 2. Basic Quantum Mechanics				
Course Summary	The Quantum Computation and Quantum Information course provides students with a comprehensive understanding of quantum computing and quantum information theory. Fundamental principles including superposition, entanglement, and quantum gates are explored, laying the groundwork for quantum computation. Students delve into advanced quantum algorithms such as Shor's and Grover's algorithms, which promise exponential speedup over classical counterparts for specific tasks. Additionally, the course examines practical applications like quantum teleportation, super dense coding, quantum error correction, and quantum key distribution, showcasing the real-world implications of quantum information processing. By the end of the course, students emerge equipped with both theoretical knowledge and practical insights, positioning them at the forefront of this rapidly evolving field.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Comprehensive Understanding of Mathematical Formulations of Quantum Mechanics.	U	C	Viva Voce/ Seminar / Quiz
CO2	Proficiency in Analyzing and Utilizing Entanglement.	An	P	Practical Assignment / Group Discussion
CO3	Mastery of Quantum Gates and Circuits.	Ap	P	Seminar Presentation / Group Tutorial Work
CO4	Application of Entanglement and Management of Quantum Noise.	An	P	Instructor-created exams / Home Assignments
CO5	Proficient Use of Tools in Quantum Information Theory.	Ap	M	Viva Voce
CO6	Integration and Application of Quantum Concepts in Practical Scenarios.	C	M	Group Discussion/ Quiz
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (12+48)	Marks (70)
I		OPEN ENDED MODULE: MATHEMATICAL FORMULATIONS OF QUANTUM MECHANICS	12	

II	INTRODUCTION TO INFORMATION THEORY		14	20
	1	Classical information, Information content in a signal, Entropy and Shannon's information theory, Probability basics (Chapter 1)	2	
	2	Representing composite states in Quantum mechanics, Tensor products of column vectors, Operators and tensor products, Tensor products of matrices (Chapter 4)	3	
	3	The Density Operator for a Pure State, Time Evolution of the Density Operator, The Density Operator for a Mixed State, Key Properties of a Density Operator (Chapter 5)	2	
	4	Probability of Obtaining a Given Measurement Result, Characterizing Mixed States, Probability of Finding an Element of the Ensemble in a Given State (Chapter 5)	2	
	5	The Partial Trace and the Reduced Density Operator, The Density Operator and the Bloch Vector (Chapter 5)	2	
	6	Distinguishing Quantum States and Measurement, Projective Measurements (Chapter 6)	1	
	7	Measurements on Composite Systems, Generalized Measurements, Positive Operator-Valued Measures (Chapter 6)	2	
	Sections from References: Chapters – 1, 4, 5, 6 of Book 1.			
III	ENTANGLEMENT, QUANTUM GATE AND CIRCUITS		18	25
	8	Bell's Theorem, Bipartite system and the Bell basis, When is a state entangled (Chapter 7)	2	
	9	The Pauli Representation, Entanglement Fidelity, Using Bell States for Density Operator Representation (Chapter 7)	2	
	10	Schmidt Decomposition, Purification (Chapter 7)	2	
	11	Classical logical gates, Single qubit gates, More single qubit gates (Chapter 8)	2	

	12	Basic quantum circuit diagram, Controlled gates, Gate decomposition (Chapter 8)	2	
	13	Hadamard gates, The phase gate. (Chapter 9)	1	
	14	Quantum Interference, Quantum Parallelism, Deutsch Jozsa Algorithm (Chapter 9)	3	
	15	Quantum Fourier transform, Shor's Algorithm, Quantum Searching and Grover's Algorithm (Chapter 9)	4	
Sections from References: Chapters – 7, 8, 9 of Book 1.				
IV	APPLICATION OF ENTANGLEMENT AND QUANTUM NOISE		11	18
	16	Teleportation, Entanglement swapping, Superdense coding (Chapter 10)	3	
	17	A brief overview of RSA encryption, Basic quantum cryptography (Chapter 10)	2	
	18	The B92 protocol, The B91 protocol (Chapter 11)	2	
	19	Single qubit errors, Quantum Operation and Krauss operators (Chapter 12)	2	
	20	The depolarization channel, The bit flip and phase flip channels, Amplitude damping, Phase damping, Quantum error correction. (Chapter 12)	2	
Sections from References: Chapters – 10, 11, 12 of Book 1.				
V	TOOLS OF QUANTUM INFORMATION THEORY		5	7
	21	The no-cloning theorem, Trace distance, Fidelity (Chapter 13)	2	
	22	Entanglement of formation and concurrence, Information content and entropy. (Chapter 13)	3	
Sections from References: Chapter – 13 of Book 1.				

Books and References:

1. Quantum Computing Explained – David McMahon (Book 1)
2. Quantum Computation and Quantum Information – Michael A Nielsen and Isaac L Chuang (Book 2)

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO 4	PSO 5	PS O6	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7
CO 1	3	0	0	2	0	0	3	2	0	2	0	0	2
CO 2	0	2	0	2	0	0	3	2	0	2	0	0	2
CO 3	0	0	2	2	0	0	3	2	0	2	0	0	1
CO 4	0	0	0	2	0	2	3	2	0	2	0	0	1
CO 5	0	0	0	2	2	0	3	2	0	2	0	0	2
CO 6	0	0	2	0	0	3	3	2	0	2	0	0	1

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory/ Practical Exam	Assignment t /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	